

Fault setting condition(s)

- For **DTC-DFCC** 00EE77:
 - Change in engine coolant temperature of less than 2 °C (35.6 °F) at diagnostic conditions
- For **DTC-DFCC** 00EE78:
 - The measured engine coolant temperature is lower than the model temperature at diagnostic conditions

P0117

Diagnostic information - DME (DFI) control unit

Engine coolant temperature sensor: electrical fault

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Engine coolant temperature sensor faulty

Fault setting condition(s)

- Engine coolant temperature (measured) < -40 °C (-40 °F) (P0117), with:
 - Ignition on or engine running
 - and
 - Intake air temperature > -30 °C (-22 °F)
- or
- Engine running > 300 second(s)
- or
- Engine coolant temperature (measured) > +130 °C (266 °F) (P0118), with:
 - Ignition on or engine running

P0118

Diagnostic information - DME (DFI) control unit

Engine coolant temperature sensor: electrical fault

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Engine coolant temperature sensor faulty

Fault setting condition(s)

- Engine coolant temperature (measured) < -40 °C (-40 °F) (P0117), with:
 - Ignition on or engine running
 - and
 - Intake air temperature > -30 °C (-22 °F)or
- Engine running > 300 second(s)
- or
- Engine coolant temperature (measured) > +130 °C (266 °F) (P0118), with:
 - Ignition on or engine running

P0119

Diagnostic information - DME (DFI) control unit

Engine coolant temperature sensor: loose contact

Diagnostic conditions

- Ignition on
- None of the following faults stored: P0117, P0118

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Engine coolant temperature sensor faulty

Fault setting condition(s)

- Engine coolant temperature: Actual-value jumps > 3 °C (37 °F)

Fault effects

- ♦ Substitute value = last actual value that was detected as OK, with:
 - Actual-value jumps > 9.75 °C (49.5 °F)

Diagnostic information - DME (DFI) control unit

Accelerator pedal potentiometer implausible

Diagnostic conditions

- Ignition on > 3 second(s)
- then accelerator pedal pressed down fully > 3 second(s)
- then release accelerator pedal fully
- then idling > 10 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in wiring to potentiometer 1 or 2
- ♦ Accelerator pedal potentiometer 1 or 2 faulty

Fault setting condition(s)

- The two potentiometer values differ too greatly from each other.
- It is not clear which value is incorrect.

Fault effects

- ♦ Engine does not respond to throttle, engine speed 1,000 rpm

P0122

Diagnostic information - DME (DFI) control unit

Accelerator pedal potentiometer 1

Diagnostic conditions

- Ignition on > 3 second(s)
- then accelerator pedal pressed down fully > 3 second(s)
- then release accelerator pedal fully
- then idling > 10 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in wiring to potentiometer 1
- ♦ Accelerator pedal potentiometer 1 faulty

Fault effects

- ♦ The accelerator pedal angle is calculated from the remaining potentiometer signal.
- ♦ No acceleration initially + increased idle speed (approx. 800 rpm)
 - ▶ until the accelerator pedal is released fully.
- ♦ Then acceleration, with
 - ♦ significantly reduced engine torque (max. approx. 100 ... 180 Nm)
 - ♦ and limited engine speed (max. approx. 5,000 rpm)
- ♦ If the accelerator and brake are pressed at the same time, the engine will run at idle speed.

P0123

Diagnostic information - DME (DFI) control unit

Accelerator pedal potentiometer 1

Diagnostic conditions

- Ignition on > 3 second(s)
- then accelerator pedal pressed down fully > 3 second(s)
- then release accelerator pedal fully
- then idling > 10 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground/between lines or open circuit in wiring to potentiometer 1
- ♦ Accelerator pedal potentiometer 1 faulty

Fault effects

- ♦ The accelerator pedal angle is calculated from the remaining potentiometer signal.
- ♦ No acceleration initially + increased idle speed (approx. 800 rpm)
 - ▶ until the accelerator pedal is released fully.
- ♦ Then acceleration, with
 - ♦ significantly reduced engine torque (max. approx. 100 ... 180 Nm)
 - ♦ and limited engine speed (max. approx. 5,000 rpm)
- ♦ If the accelerator and brake are pressed at the same time, the engine will run at idle speed.

P0131

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 1

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

♦ Do not interchange the oxygen sensor upstream of the catalytic converter with the sensor downstream of the catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Idling > 3 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Oxygen sensor upstream of catalytic converter bank 1 faulty

P0132

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 1

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

♦ Do not interchange the oxygen sensor upstream of the catalytic converter with the sensor downstream of the catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Idling > 3 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Oxygen sensor upstream of catalytic converter bank 1 faulty

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 1, dynamic response

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

- ♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.
- ♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence (without tank leak test).

or

- Engine running > 2 minute(s) with:
 - Engine at operating temperature
 - All oxygen sensor(s) ready for operation
- Then steady driving > 10 minute(s) at:
 - Engine speed < 800 ... 3,200 rpm
 - Mass air flow 120 ... 520 mg/stroke
 - No overrun phase(s)

Possible fault causes

- ♦ Oxygen sensor upstream of catalytic converter, bank 1 faulty (aged)

P0135

Diagnostic information - DME (DFI) control unit

Oxygen sensor heater circuit upstream of catalytic converter, bank 1

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Engine running, with:
 - **Oxygen sensor heater** downstream of catalytic converter active > 180 second(s)
 - Lambda control active
- None of the following faults stored: P0030, P0031, P0032

Possible fault causes

- ♦ Contact resistance (power supply or activation)
- ♦ Shunt (high resistance) in heater circuit ⇒ Signal line
- ♦ Oxygen sensor upstream of catalytic converter bank 1 faulty

P0136

Diagnostic information - DME (DFI) control unit

Oxygen sensor downstream of catalytic converter, bank 1

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

- ♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.
- ♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence (without tank leak test).

or

- Drive > 2 minute(s) with:
 - Engine at operating temperature
 - Mass air flow > 10 kg/h
- Then idling > 30 second(s)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Oxygen sensor downstream of catalytic converter, bank 1 faulty

P0137

Diagnostic information - DME (DFI) control unit

Oxygen sensor downstream of catalytic converter, bank 1

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

- ♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.
- ♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence (without tank leak test).

or

- Drive > 2 minute(s) with:
 - Engine at operating temperature
 - Mass air flow > 10 kg/h
- Then idling > 30 second(s)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Oxygen sensor downstream of catalytic converter, bank 1 faulty

P0138

Diagnostic information - DME (DFI) control unit

Oxygen sensor downstream of catalytic converter, bank 1

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

- ♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.
- ♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence (without tank leak test).

or

- Drive > 2 minute(s) with:
 - Engine at operating temperature
 - Mass air flow > 10 kg/h
- Then idling > 30 second(s)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Oxygen sensor downstream of catalytic converter, bank 1 faulty

P013A

Diagnostic information - DME (DFI) control unit

Oxygen sensor downstream of catalytic converter, bank 1

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence

or

- Engine temperature > 60 °C (140 °F)
- Time since engine start > 10 second(s)
- Catalytic converter temperature 540 ... 750 °C (1,004 ... 1,382 °C)
- Drive for 60 seconds with slight acceleration at 40 ... 130 km/h (24 ... 80 mph)
- Active carbon filter loading < 0.7
- Engine speed < 3,000 rpm
- Lambda control ahead of catalytic converter active
- Lambda control behind catalytic converter active
- None of the following faults stored: Oxygen sensor(s) - electrical fault, intake manifold pressure sensor or throttle valve adjusting unit

Possible fault causes

♦ Oxygen sensor downstream of catalytic converter, bank 1 faulty

P013B

Diagnostic information - DME (DFI) control unit

Oxygen sensor downstream of catalytic converter, bank 1

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence

or

- Engine temperature > 60 °C (140 °F)
- Time since engine start > 10 second(s)
- Catalytic converter temperature 540 ... 750 °C (1,004 ... 1,382 °C)
- Drive for 60 seconds with slight acceleration at 40 ... 130 km/h (24 ... 80 mph)
- Active carbon filter loading < 0.7
- Engine speed < 3,000 rpm
- Lambda control ahead of catalytic converter active
- Lambda control behind catalytic converter active
- None of the following faults stored: Oxygen sensor(s) - electrical fault, intake manifold pressure sensor or throttle valve adjusting unit

Possible fault causes

♦ Oxygen sensor downstream of catalytic converter, bank 1 faulty

P013C

Diagnostic information - DME (DFI) control unit

Oxygen sensor downstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence

or

- Engine temperature > 60 °C (140 °F)
- Time since engine start > 10 second(s)
- Catalytic converter temperature 540 ... 750 °C (1,004 ... 1,382 °F)
- Drive for 60 seconds with slight acceleration at 40 ... 130 km/h (24 ... 80 mph)
- Active carbon filter loading < 0.7
- Engine speed < 3,000 rpm
- Lambda control ahead of catalytic converter active
- Lambda control behind catalytic converter active
- None of the following faults stored: Oxygen sensor(s) - electrical fault, intake manifold pressure sensor or throttle valve adjusting unit

Possible fault causes

- ♦ Oxygen sensor downstream of catalytic converter, bank 2 faulty

P013D

Diagnostic information - DME (DFI) control unit

Oxygen sensor downstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence

or

- Engine temperature > 60 °C (140 °F)
- Time since engine start > 10 second(s)
- Catalytic converter temperature 540 ... 750 °C (1,004 ... 1,382 °F)
- Drive for 60 seconds with slight acceleration at 40 ... 130 km/h (24 ... 80 mph)
- Active carbon filter loading < 0.7
- Engine speed < 3,000 rpm
- Lambda control ahead of catalytic converter active
- Lambda control behind catalytic converter active
- None of the following faults stored: Oxygen sensor(s) - electrical fault, intake manifold pressure sensor or throttle valve adjusting unit

Possible fault causes

♦ Oxygen sensor downstream of catalytic converter, bank 2 faulty

Diagnostic information - DME (DFI) control unit

Oxygen sensor downstream of catalytic converter, bank 1

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence (without tank leak test).

or

- Engine temperature > 60 °C (140 °F)
- Time since engine start > 10 second(s)
- Catalytic converter temperature 540 ... 750 °C (1,004 ... 1,382 °F)
- Drive for 60 seconds with slight acceleration at 40 ... 130 km/h (24 ... 80 mph)
- Active carbon filter loading < 0.7
- Engine speed < 3,000 rpm
- Lambda control ahead of catalytic converter active
- Lambda control behind catalytic converter active
- None of the following faults stored: Oxygen sensor(s) - electrical fault, intake manifold pressure sensor or throttle valve adjusting unit

Possible fault causes

♦ Oxygen sensor downstream of catalytic converter, bank 1 faulty

Fault setting condition(s)

- Excessive time offset during edge changes between the lean and rich cycle or between the rich and lean cycle.

Diagnostic information - DME (DFI) control unit

Oxygen sensor downstream of catalytic converter, bank 1

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence (without tank leak test).

or

- Engine temperature > 60 °C (140 °F)
- Time since engine start > 10 second(s)
- Catalytic converter temperature 540 ... 750 °C (1,004 ... 1,382 °F)
- Drive for 60 seconds with slight acceleration at 40 ... 130 km/h (24 ... 80 mph)
- Active carbon filter loading < 0.7
- Engine speed < 3,000 rpm
- Lambda control ahead of catalytic converter active
- Lambda control behind catalytic converter active
- None of the following faults stored: Oxygen sensor(s) - electrical fault, intake manifold pressure sensor or throttle valve adjusting unit

Possible fault causes

♦ Oxygen sensor downstream of catalytic converter, bank 1 faulty

Fault setting condition(s)

- Excessive time offset during edge changes between the lean and rich cycle or between the rich and lean cycle.

Diagnostic information - DME (DFI) control unit

Oxygen sensor heater circuit downstream of catalytic converter, bank 1

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

- ♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.
- ♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence (without tank leak test).
- or
- Engine running, with:
 - **Oxygen sensor heater** downstream of catalytic converter active > 180 second(s)
 - Lambda control active
 - None of the following faults stored: P0036, P0037, P0038

Possible fault causes

- ♦ Contact resistance (power supply or activation)
- ♦ Shunt (high resistance) in heater circuit ⇒ signal line
- ♦ Oxygen sensor downstream of catalytic converter, bank 1 faulty

P014A

Diagnostic information - DME (DFI) control unit

Oxygen sensor downstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

• Perform all short tests in sequence

or

- Engine temperature > 60 °C (140 °F)
- Time since engine start > 10 second(s)
- Catalytic converter temperature 540 ... 750 °C (1,004 ... 1,382 °F)
- Drive for 60 seconds with slight acceleration at 40 ... 130 km/h (24 ... 80 mph)
- Active carbon filter loading < 0.7
- Engine speed < 3,000 rpm
- Lambda control ahead of catalytic converter active
- Lambda control behind catalytic converter active
- None of the following faults stored: Oxygen sensor(s) - electrical fault, intake manifold pressure sensor or throttle valve adjusting unit

Possible fault causes

♦ Oxygen sensor downstream of catalytic converter, bank 1 faulty

Fault setting condition(s)

• Excessive time offset during edge changes between the lean and rich cycle or between the rich and lean cycle.

P014B

Diagnostic information - DME (DFI) control unit

Oxygen sensor downstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

• Perform all short tests in sequence

or

- Engine temperature > 60 °C (140 °F)
- Time since engine start > 10 second(s)
- Catalytic converter temperature 540 ... 750 °C (1,004 ... 1,382 °F)
- Drive for 60 seconds with slight acceleration at 40 ... 130 km/h (24 ... 80 mph)
- Active carbon filter loading < 0.7
- Engine speed < 3,000 rpm
- Lambda control ahead of catalytic converter active
- Lambda control behind catalytic converter active
- None of the following faults stored: Oxygen sensor(s) - electrical fault, intake manifold pressure sensor or throttle valve adjusting unit

Possible fault causes

♦ Oxygen sensor downstream of catalytic converter, bank 1 faulty

Fault setting condition(s)

• Excessive time offset during edge changes between the lean and rich cycle or between the rich and lean cycle.

P0151

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange the oxygen sensor upstream of the catalytic converter with the sensor downstream of the catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Idling > 3 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Oxygen sensor upstream of catalytic converter bank 2 faulty

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange the oxygen sensor upstream of the catalytic converter with the sensor downstream of the catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Idling > 3 second(s)

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Oxygen sensor upstream of catalytic converter bank 2 faulty

Diagnostic information - DME (DFI) control unit

Oxygen sensor upstream of catalytic converter, bank 2, dynamic response

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence (without tank leak test).

or

- Engine running > 2 minute(s) with:
 - Engine at operating temperature
 - All oxygen sensor(s) ready for operation
- Then steady driving > 10 minute(s) at:
 - Engine speed < 800 ... 3,200 rpm
 - Mass air flow 120 ... 520 mg/stroke
 - No overrun phase(s)

Possible fault causes

♦ Oxygen sensor upstream of catalytic converter, bank 2 faulty (aged)

Diagnostic information - DME (DFI) control unit

Oxygen sensor heater circuit upstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence (without tank leak test).

or

- Engine running, with:

- **Oxygen sensor heater** downstream of catalytic converter active > 180 second(s)

- Lambda control active

- None of the following faults stored: P0050, P0051, P0052

Possible fault causes

- ♦ Contact resistance (power supply or activation)

- ♦ Shunt (high resistance) in heater circuit ⇒ Signal line

- ♦ Oxygen sensor upstream of catalytic converter bank 2 faulty

P0156

Diagnostic information - DME (DFI) control unit

Oxygen sensor downstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

• Perform all short tests in sequence (without tank leak test).

or

- Drive > 2 minute(s) with:
 - Engine at operating temperature
 - Mass air flow > 10 kg/h
- Then idling > 30 second(s)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Oxygen sensor downstream of catalytic converter, bank 2 faulty

P0157

Diagnostic information - DME (DFI) control unit

Oxygen sensor downstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence (without tank leak test).

or

- Drive > 2 minute(s) with:
 - Engine at operating temperature
 - Mass air flow > 10 kg/h
- Then idling > 30 second(s)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Oxygen sensor downstream of catalytic converter, bank 2 faulty

P0158

Diagnostic information - DME (DFI) control unit

Oxygen sensor downstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

• Perform all short tests in sequence (without tank leak test).

or

- Drive > 2 minute(s) with:
 - Engine at operating temperature
 - Mass air flow > 10 kg/h
- Then idling > 30 second(s)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Oxygen sensor downstream of catalytic converter, bank 2 faulty

P0161

Diagnostic information - DME (DFI) control unit

Oxygen sensor heater circuit downstream of catalytic converter, bank 2

NOTE

Plug connection disconnected = reference-air duct open!

Damage to oxygen sensor!

♦ **DO NOT** use spray, grease, fluid or similar products on the oxygen sensor plug connections.

i INFORMATION

♦ Do not interchange oxygen sensor upstream of catalytic converter with oxygen sensor downstream of catalytic converter as this will result in implausible fault entries.

♦ The connectors themselves are coded and cannot be interchanged.

Diagnostic conditions

- Perform all short tests in sequence (without tank leak test).

or

- Engine running, with:

- **Oxygen sensor heater** downstream of catalytic converter active > 180 second(s)

- Lambda control active

- None of the following faults stored: P0056, P0057, P0058

Possible fault causes

- ♦ Contact resistance (power supply or activation)

- ♦ Shunt (high resistance) in heater circuit ⇒ signal line

- ♦ Oxygen sensor downstream of catalytic converter, bank 2 faulty

Diagnostic information - DME (DFI) control unit

Fuel high pressure adaptation limit

INFORMATION 1

- ♦ Faults in the fuel high-pressure injection system area are frequently caused by defects in the pre-delivery circuit (fuel low pressure).
- ♦ Therefore, the pre-delivery circuit must be checked first in case of problems in the area of the fuel high-pressure injection system.

Diagnostic conditions

- Engine speed 450 ... 4,000 rpm
- Engine temperature - 10 ... + 100 °C (14 ... 212 °F)
- Driving with different operating conditions

Possible fault causes

- ♦ Pre-delivery quantity or pressure in the fuel low-pressure circuit too low
- ♦ Learned adaptation values do not match the installed fuel high-pressure pump, e.g. after replacing the pump or engine
 - ⇒ Fuel high pressure adaptation required ► see  2
- ♦ Fuel high-pressure pump faulty

INFORMATION 2

The old values must be deleted prior to a new adaptation of the fuel high-pressure ranges.

- ♦ The fuel high-pressure adaptation values can be deleted using the PIWIS Tester as follows:
DME ⇒ Maintenance/repairs ⇒ Delete adaptation values
- ♦ Adaptation takes place while driving.
- ♦ The system test can be carried out in order to test the pump:
DME ⇒ Maintenance/repairs ⇒ System tests ⇒ High-pressure pump

P016F

Diagnostic information - DME (DFI) control unit

Fuel high pressure adaptation limit

INFORMATION 1

- ♦ Faults in the fuel high-pressure injection system area are frequently caused by defects in the pre-delivery circuit (fuel low pressure).
- ♦ Therefore, the pre-delivery circuit must be checked first in case of problems in the area of the fuel high-pressure injection system.

Diagnostic conditions

- Engine speed 450 ... 4,000 rpm
- Engine temperature - 10 ... + 100 °C (14 ... 212 °F)
- Driving with different operating conditions

Possible fault causes

- ♦ Pre-delivery quantity or pressure in the fuel low-pressure circuit too low
- ♦ Learned adaptation values do not match the installed fuel high-pressure pump, e.g. after replacing the pump or engine
 - ⇒ Fuel high pressure adaptation required ► see  2
- ♦ Fuel high-pressure pump faulty

INFORMATION 2

The old values must be deleted prior to a new adaptation of the fuel high-pressure ranges.

- ♦ The fuel high-pressure adaptation values can be deleted using the PIWIS Tester as follows:
DME ⇒ Maintenance/repairs ⇒ Delete adaptation values
- ♦ Adaptation takes place while driving.
- ♦ The system test can be carried out in order to test the pump:
DME ⇒ Maintenance/repairs ⇒ System tests ⇒ High-pressure pump

P0171

Diagnostic information - DME (DFI) control unit

Lambda controller or lambda control adaptation, bank 1: system too lean

Diagnostic conditions

- Short test

or

- Lambda control active
- Lambda control adapted
- Engine temperature > 60 °C (140 °F)
- Intake air temperature < 80 °C (176 °F)
- Engine running > 45 second(s), at:
 - Engine speed < 800 rpm
- Then drive with different operating conditions in the lower and upper load ranges with valve lift switchover from small lift to large lift ([A220](#)).

Possible fault causes

- ♦ Leak in exhaust system (engine ⇒ oxygen sensor upstream of catalytic converter)
- ♦ Exhaust system constricted/blocked
- ♦ Fuel injector faulty (mechanical fault)
- ♦ Timing set incorrectly
- ♦ Fault in areas: fuel high pressure, intake manifold pressure sensor, throttle valve adjusting unit, ambient pressure sensor, intake air temperature sensor, rpm sensor, valve lift control, camshaft adjustment

Fault setting condition(s)

- P0171 - 30% deviation of the lambda controller for more than 30 seconds
- P2187 - 3.5 mg/stroke deviation in lambda control adaptation in the range near idle speed
- P2177 - 30% deviation in lambda control adaptation in the lower partial load range

INFORMATION

Opposed adaptation values in connection with misfires indicate that the timing is out of adjustment.

P0172

Diagnostic information - DME (DFI) control unit

Lambda controller or lambda control adaptation, bank 1: system too rich

Diagnostic conditions

- Short test

or

- Lambda control active
- Lambda control adapted
- Engine temperature > 60 °C (140 °F)
- Intake air temperature < 80 °C (176 °F)
- Engine running > 45 second(s), at:
 - Engine speed < 800 rpm
- Then drive with different operating conditions in the lower and upper load ranges with valve lift switchover from small lift to large lift ([A220](#)).

Possible fault causes

- ♦ High fuel content in engine oil.
- ♦ Leak in exhaust system (engine ⇒ oxygen sensor upstream of catalytic converter)
- ♦ Fuel injector faulty (mechanical fault)
- ♦ Tank vent valve stuck open
- ♦ Timing set incorrectly
- ♦ Fault in areas: fuel high pressure, intake manifold pressure sensor, throttle valve adjusting unit, ambient pressure sensor, intake air temperature sensor, rpm sensor, valve lift control, camshaft adjustment

Fault setting condition(s)

- P0172 - 30% deviation of the lambda controller for more than 30 seconds
- P2188 - 3.5 mg/stroke deviation in lambda control adaptation in the range near idle speed
- P2178 - 30% deviation in lambda control adaptation in the lower partial load range

INFORMATION

Opposed adaptation values in connection with misfires indicate that the timing is out of adjustment.

P0174

Diagnostic information - DME (DFI) control unit

Lambda controller or lambda control adaptation, bank 2: system too lean

Diagnostic conditions

- Short test

or

- Lambda control active
- Lambda control adapted
- Engine temperature > 60 °C (140 °F)
- Intake air temperature < 80 °C (176 °F)
- Engine running > 45 second(s), at:
 - Engine speed < 800 rpm
- Then drive with different operating conditions in the lower and upper load ranges with valve lift switchover from small lift to large lift ([A220](#)).

Possible fault causes

- ♦ Leak in exhaust system (engine ⇒ oxygen sensor upstream of catalytic converter)
- ♦ Exhaust system constricted/blocked
- ♦ Fuel injector faulty (mechanical fault)
- ♦ Timing set incorrectly
- ♦ Fault in areas: fuel high pressure, intake manifold pressure sensor, throttle valve adjusting unit, ambient pressure sensor, intake air temperature sensor, rpm sensor, valve lift control, camshaft adjustment

Fault setting condition(s)

- P0174 - 30% deviation of the lambda controller for more than 30 seconds
- P2189 - 3.5 mg/stroke deviation in lambda control adaptation in the range near idle speed
- P2179 - 30% deviation in lambda control adaptation in the lower partial load range

INFORMATION

Opposed adaptation values in connection with misfires indicate that the timing is out of adjustment.

P0175

Diagnostic information - DME (DFI) control unit

Lambda controller or lambda control adaptation, bank 2: system too rich

Diagnostic conditions

- Short test

or

- Lambda control active
- Lambda control adapted
- Engine temperature > 60 °C (140 °F)
- Intake air temperature < 80 °C (176 °F)
- Engine running > 45 second(s), at:
 - Engine speed < 800 rpm
- Then drive with different operating conditions in the lower and upper load ranges with valve lift switchover from small lift to large lift ([A220](#)).

Possible fault causes

- ♦ High fuel content in engine oil.
- ♦ Leak in exhaust system (engine ⇒ oxygen sensor upstream of catalytic converter)
- ♦ Fuel injector faulty (mechanical fault)
- ♦ Tank vent valve stuck open
- ♦ Timing set incorrectly
- ♦ Fault in areas: fuel high pressure, intake manifold pressure sensor, throttle valve adjusting unit, ambient pressure sensor, intake air temperature sensor, rpm sensor, valve lift control, camshaft adjustment

Fault setting condition(s)

- P0175 - 30% deviation of the lambda controller for more than 30 seconds
- P2190 - 3.5 mg/stroke deviation in lambda control adaptation in the range near idle speed
- P2180 - 30% deviation in lambda control adaptation in the lower partial load range

INFORMATION

Opposed adaptation values in connection with misfires indicate that the timing is out of adjustment.

P0190

Diagnostic information - DME (DFI) control unit

Fuel high-pressure sensor

Diagnostic conditions

- Time since engine start > 5 second(s)

Possible fault causes

- ♦ Loose contact or corrosion on connectors or lines
- ♦ Short circuit to B+/ground/between lines or open circuit in the lines
- ♦ Fuel high-pressure sensor faulty

Diagnostic information - DME (DFI) control unit

Fuel high-pressure sensor implausible

INFORMATION

- ♦ Faults in the fuel high-pressure injection system area are frequently caused by defects in the pre-delivery circuit (fuel low pressure).
- ♦ Therefore, the pre-delivery circuit must be checked first in case of problems in the area of the fuel high-pressure injection system.

Diagnostic conditions

- Time since engine start > 250 second(s)
- Engine temperature - 10 ... + 100 °C (14 ... 212 °F)
- Engine speed 450 ... 4,000 rpm
- Lambda control active

Possible fault causes

 The value **DTC-DFCC** is shown in the extended fault memory for the respective fault code.

- For **DTC-DFCC** 00F046:
 - ♦ Pre-delivery quantity or pressure in the fuel low-pressure circuit too low
 - ♦ Fuel high-pressure sensor faulty
- For **DTC-DFCC** 00F040 or 00F041:
 - ♦ Pre-delivery quantity or pressure in the fuel low-pressure circuit too low
 - ♦ Fuel high-pressure sensor faulty
 - ♦ Leak in fuel high-pressure system
 - ♦ Fuel high-pressure sensor faulty
 - ♦ Fuel high-pressure pump faulty
 - ♦ Insufficient fuel due to foreign bodies (chips) ahead of the fuel high-pressure rail (see  below)

Fault setting condition(s)

- For **DTC-DFCC** 00F046:
 - No change in the fuel high-pressure sensor voltage line with engine running
- For **DTC-DFCC** 00F040 or 00F041:
 - The fuel high-pressure controller is compared with the reaction of the lambda controller. If the values behave implausibly with respect to each other, a test is also performed to check whether the fuel high-pressure actual value deviates significantly from the setpoint value. If it does, the fault is entered.
 - Example: The fuel high-pressure controller increases the pressure because the fuel high-pressure sensor measures pressure that is too low for the engine's operating state. As a result, the lambda control system detects a mixture that is too rich.

INFORMATION

Example

- ♦ Vehicle speed approx. 140 km/h (86 mph)
- ♦ Little or no acceleration although accelerator pedal is floored
- ♦ Actual values, or values in extended fault memory when fault was entered or when lack of power was detected, showed:
 - ♦ Fault entry P2414
 - ♦ Oxygen sensing, bank 1 = 25.0% (enrichment)
 - ♦ Fuel trim adaptation (load-dependent), bank 1 = 14.844%
 - ♦ L070_Oxygen sensor (LSU) voltage ahead of cat. con. 1 = 2.98828 V
 - ♦ L090_Oxygen sensor voltage behind cat. con. 1 = 0.00000 V

Explanation

The inlet for the fuel high-pressure rail behind the high-pressure pump is blocked either fully or partially by foreign bodies.

OR

The connection between the two fuel high-pressure rails is blocked either fully or partially by foreign bodies.

 The inlet of the respective rail is constricted by a restrictor, which is used to dampen pressure peaks.

- ♦ The fuel high-pressure sensor, which is fitted at the end of the lines, measures a pressure increase that is too slow, or a rapid drop in pressure when the fuel injectors open.
- ♦ As a result, fuel high pressure that is too high is set in the rail AHEAD OF the blockage, while at the same time, there is insufficient pressure and volume BEHIND the blockage.
- ♦ The oxygen sensors of the affected bank measure a large oxygen surplus, while at the same time, fuel trim adaptation tries to enrich the mixture as much as possible (extension of the injection time).

Consequences

- ♦ The following faults can be entered individually, in combination, or might not be entered at all:
 - ♦ Mixture control fault
 - ♦ Load detection fault
 - ♦ Misfiring
 - ♦ Oxygen sensor fault

AND

- ♦ Lack of power in upper load range can be reproduced, but is not necessarily always present.

OR

- ♦ The engine can stall again while starting.

Cause

After removing the lines between the fuel high-pressure pump and high-pressure rail, you can sometimes see chips or hear them rattling.